



Statistical Methods & Machine Learning 1

Class #8

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October 22, 2024

Outline

- Review
- Prediction
- Inference
 - Causation
 - Hill's Viewpoints
 - DAGs
 - Confounders, mediators, colliders, and descendents



Review

Binary Predictor Variables

```
Call:
lm(formula = wage ~ indus + info, data = wage)

Residuals:
    Min       1Q   Median       3Q      Max
-100.507  -25.362   -6.117   15.697  197.750

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  120.593      1.070   112.69  <2e-16 ***
indus        -17.272      1.492   -11.58  <2e-16 ***
info          NA         NA      NA      NA
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.83 on 2998 degrees of freedom
Multiple R-squared:  0.04281,    Adjusted R-squared:  0.04249
F-statistic: 134.1 on 1 and 2998 DF,  p-value: < 2.2e-16
```

Binary Predictor Variables

```
> head(model.matrix(mod_j1))  
      (Intercept)  induc  info  
231655          1      1     0  
86582           1      0     1  
161300          1      1     0  
155159          1      0     1  
11443           1      0     1  
376662          1      0     1
```

Both of our predictors are dummy/indicator variables derived from the `jobclass` variable.

They're simply the inverse of each other. No additional information is provided by one that's not already provided by the other. We're trying to use 3 parameters to model only 2 groups. Matrix is not of full rank.

Identifiability

$$\begin{aligned}y_i &= \mu_1 d_{1i} + \mu_2 d_{2i} + \epsilon_i \\&= (\mu_1 + \delta - \delta) d_{1i} + (\mu_2 + \delta - \delta) d_{2i} + \epsilon_i \\&= -\delta(d_{1i} + d_{2i}) + (\mu_1 + \delta) d_{1i} + (\mu_2 + \delta) d_{2i} + \epsilon_i \\&\equiv \beta_0 + \beta_1 d_{1i} + \beta_2 d_{2i} + \epsilon_i \text{ with } \epsilon_i \sim N(0, \sigma^2)\end{aligned}$$

→ $\beta_0, \beta_1, \beta_2$ not estimable; Why?

$$\text{e.g.) } \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \\ y_8 \\ y_9 \\ y_{10} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \\ \epsilon_6 \\ \epsilon_7 \\ \epsilon_8 \\ \epsilon_9 \\ \epsilon_{10} \end{pmatrix}$$

Identifiability

To put it another way, a model is said to be *nonidentifiable* if it contains parameters that cannot be estimated uniquely—or, to put it another way, that have standard errors of infinity. The offending parameters are called *nonidentified*. The most familiar and important example of nonidentifiability arises from *collinearity* (also called multicollinearity) of regression predictors. A set of predictors is collinear if there is a linear combination of them that equals 0 for all the data. We discuss this problem in the context of indicator variables at the end of Section 12.5.

A simple example of collinearity is a model predicting family outcomes given the number of boys in the family, the number of girls, and the total number of children. Labeling these predictors as x_{2i} , x_{3i} , and x_{4i} , respectively, for family i (with $x_{1i} = 1$ being reserved for the constant term in the regression), we have $x_{4i} = x_{2i} + x_{3i}$, thus $-x_{2i} - x_{3i} + x_{4i} = 0$ and the set of predictors is collinear.

There can also be problems with *near*-collinearity, which leads to poor identification. For example, suppose you try to predict people's heights from the lengths of their left and right feet. These two predictors are nearly (but not exactly) collinear, and as a result it will be difficult to untangle the two coefficients, and we can characterize the fitted regression as unstable, in the sense that if it were re-fitted with new data, a new sample from the same population, we could see much different results. This instability should be reflected in large standard errors for the estimated coefficients.

Parameters and data

In the following linear regression model what are our parameters (i.e., what are we estimating)?

- $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$

Parameters and data

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 - $\beta_0, \beta_1, \beta_2$

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- $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$
 - $\beta_0, \beta_1, \beta_2$

These parameters are what our hypothesis tests are concerned with, not X_1 or X_2 as those are our observed data.

Interaction terms

- Interaction terms allow us to vary the slope of an association between a predictor and the outcome based on the value of another predictor(s).
- Interaction terms should nearly always be accompanied in the model by its component main effects.

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- Interaction terms should nearly always be accompanied in the model by its component main effects.
 - $Y = \beta_0 + \underbrace{\beta_1 X_1 + \beta_2 X_2}_{\text{Main effects}} + \beta_3 (X_1 * X_2) + \epsilon$

Inference vs. Prediction

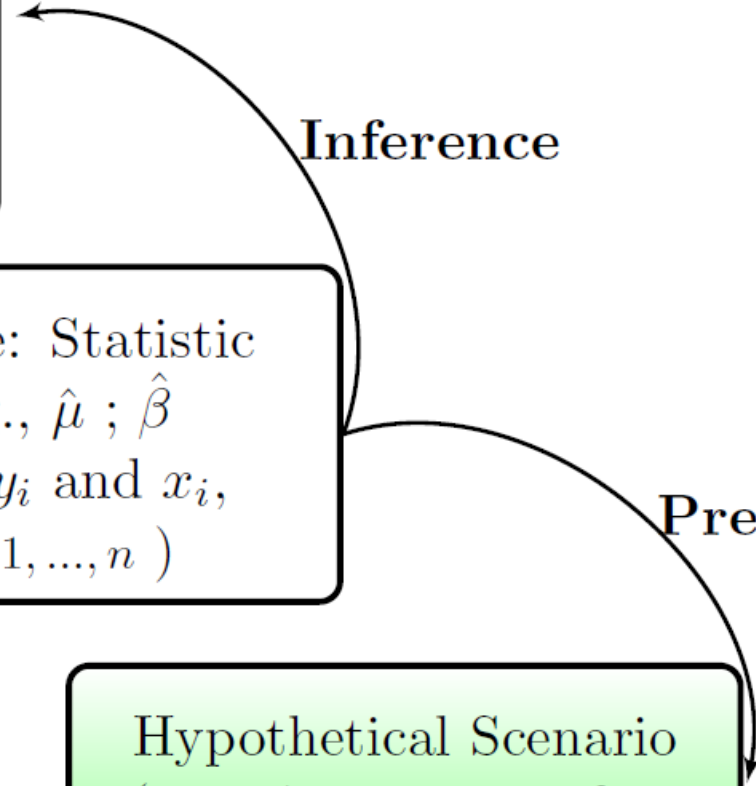
Population: Parameter
(e.g., μ : average income (y);
 β : effect of education (x) on
income (y), $y_i = \beta x_i + \epsilon_i$
from y_i and x_i , $i = 1, \dots, N$)

Sample: Statistic
(e.g., $\hat{\mu}$; $\hat{\beta}$
from y_i and x_i ,
 $i = 1, \dots, n$)

Hypothetical Scenario
(e.g., $\hat{y}_0$: income of an
individual 0 with x_0)

Inference

Prediction



Inference vs. Prediction

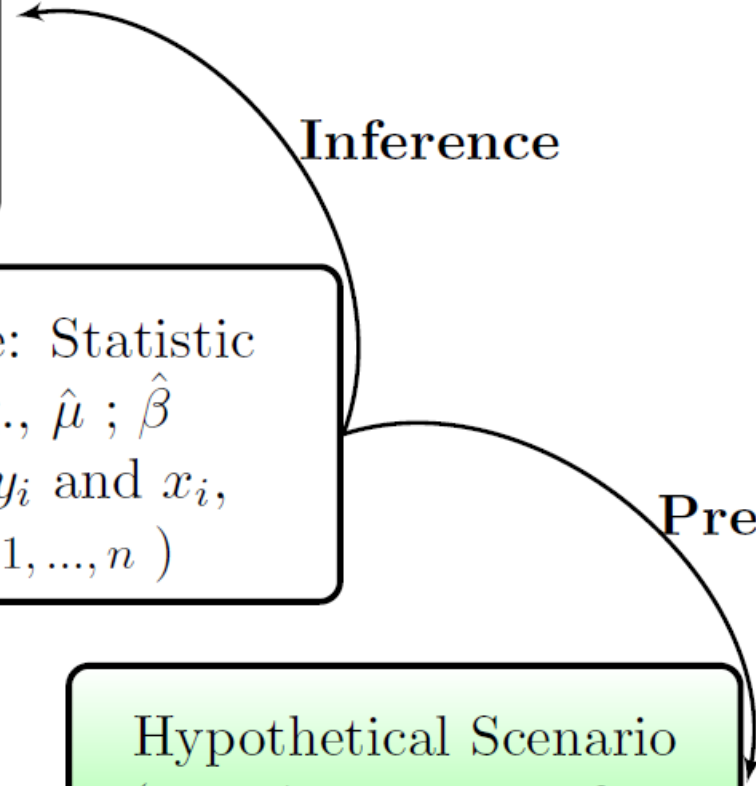
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Inference

Prediction





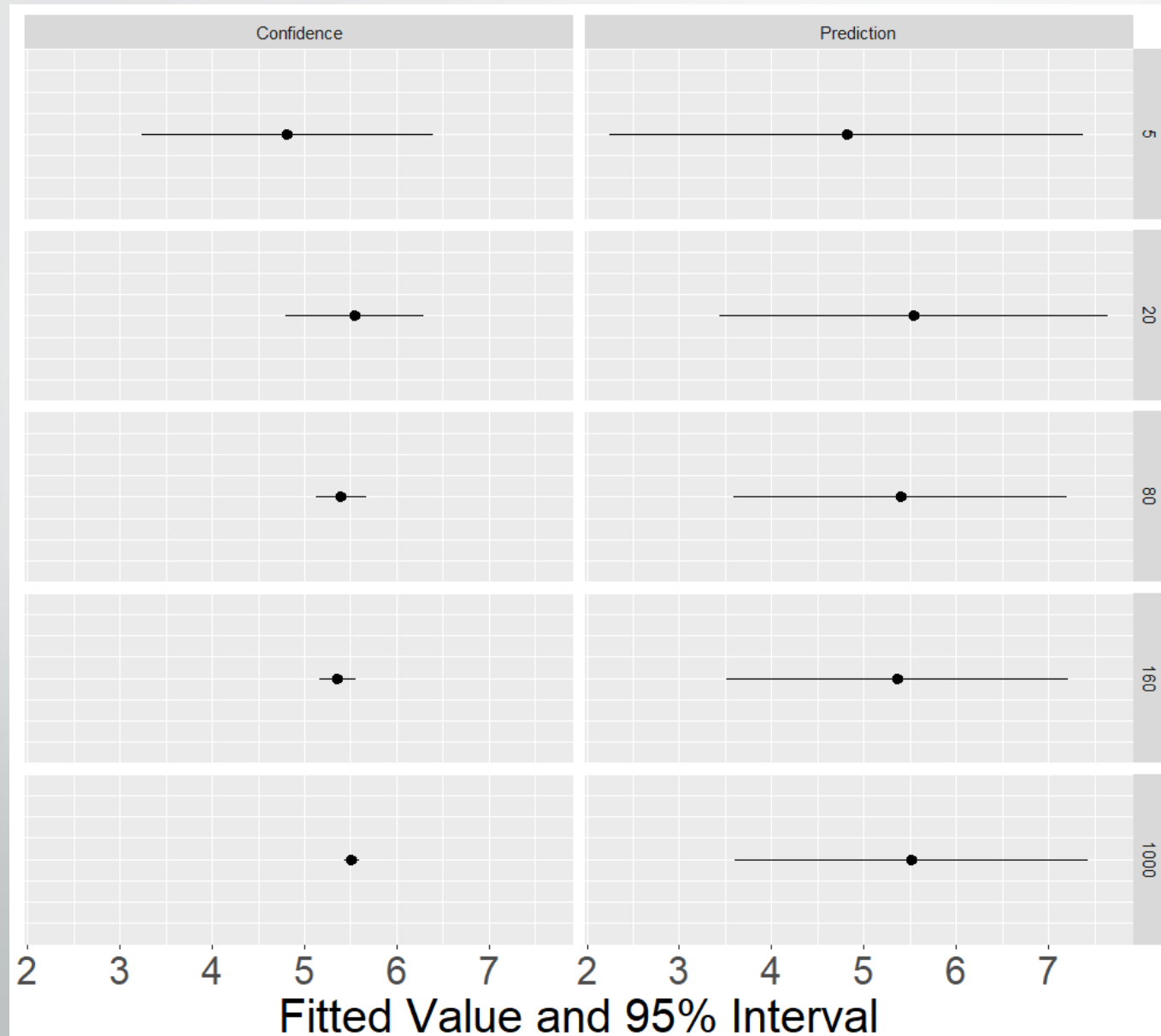
Prediction

Prediction example

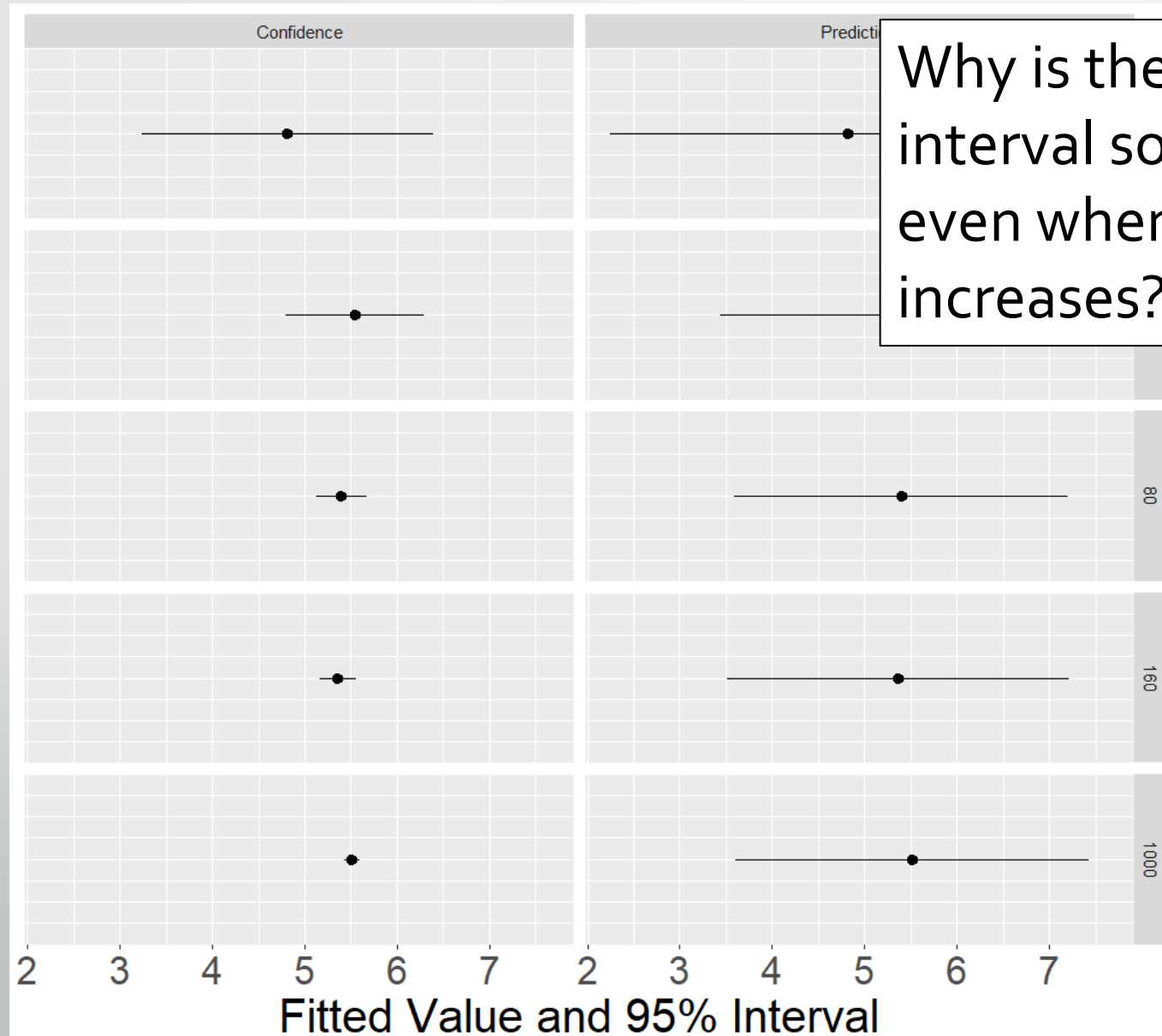
- Let's make some data
- We'll use these to fit 5 simple linear regression models with different sample size (n), estimate/predict β_1 , and calculate its 95% confidence and prediction intervals.
- What do you think will happen to each interval as sample size increases?

```
5 set.seed(894894)
6 n = 1000
7
8 df <- tibble(
9   id = 1:n,
10  x = rnorm(1000, 3.5, 1.75)
11 )
12
13 df_subsets = tibble(
14   n = c(5, 20, 80, 160, 1000),
15   data = map(k, ~ filter(df, id <= .x))
16 ) %>%
17   unnest(data)
18
19 e <- rnorm(1265, 0, 1)
20 df_subsets$y <- 1.5 + 2*df_subsets$x + e
```


Prediction example



Prediction example



Estimation of Mean/Expected Response

Best estimate of Y for a given X

From $\hat{Y}_* \sim N(\beta_0 + \beta_1 X_*, \sigma^2(n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2))$

Inference for Y_* estimation is the same as any normal distribution random variables:

$(1 - \alpha)\%$ CI of Y_* estimation: $\hat{Y}_* \pm t_{(n-2), \alpha/2} SE(\hat{Y}_*)$

Where $SE(\hat{Y}_*) = \sqrt{\sigma^2(n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2)}$

Prediction of Response

Prediction of Y for a given X

From $(Y_* - \hat{Y}_*) \sim N(0, \sigma^2(1 + n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2))$

Inferences for Y_* prediction involves error variance, unlike the inferences for Y_* estimation:

$(1 - \alpha)\%$ CI of Y_* prediction: $\hat{Y}_* \pm t_{(n-2), \alpha/2} SE(Y_* - \hat{Y}_*)$

where $SE(Y_* - \hat{Y}_*) = \sqrt{\sigma^2(1 + n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2)}$

Estimation vs. Prediction of Response

- Best estimate of Y_* for a given X

$(1 - \alpha)\%$ CI of Y_* :

$$\hat{Y}_* \pm t_{(n-2), \alpha/2} SE(\hat{Y}_*),$$

$$\text{where } SE(\hat{Y}_*) = \sqrt{\sigma^2(n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2)}$$

- Prediction of Y_* for a given X

$(1 - \alpha)\%$ CI (or Prediction Interval) of Y_* :

$$\hat{Y}_* \pm t_{(n-2), \alpha/2} SE(Y_* - \hat{Y}_*),$$

$$\text{Where } SE(Y_* - \hat{Y}_*) = \sqrt{\sigma^2(1 + n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2)}$$

- Prediction intervals are wider or narrower than confidence intervals? Why?

Estimation vs. Prediction of Response

- Best estimate of Y_* for a given X

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- Prediction intervals are wider or narrower than confidence intervals? Why?

Formula for the confidence interval and the prediction interval follow same pattern general formula:
Sample estimate $\pm$ (t -multiplier $\times$ standard error)

Estimation vs. Prediction of Response

- Best estimate of Y_* for a given X

$(1 - \alpha)\%$ CI of Y_* :

$$\hat{Y}_* \pm t_{(n-2), \alpha/2} SE(\hat{Y}_*),$$

where $SE(\hat{Y}_*) = \sqrt{\sigma^2(n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2)}$

Difference is in their standard errors as the prediction interval has to account for both error in the estimate of the model (i.e., $f(x)$) and the uncertainty regarding how much an individual point differs from the regression plane

- Prediction of Y_* for a given X

$(1 - \alpha)\%$ CI (or Prediction Interval) of Y_* :

$$\hat{Y}_* \pm t_{(n-2), \alpha/2} SE(Y_* - \hat{Y}_*),$$

Where $SE(Y_* - \hat{Y}_*) = \sqrt{\sigma^2(1 + n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2)}$

- Prediction intervals are wider or narrower than confidence intervals? Why?

Estimation vs. Prediction of Response

- Best estimate of Y_* for a given X

(1) With confidence intervals we're assessing variability/uncertainty around the mean of an estimated quantity $\bar{x}$. which as a summary measure itself, has already reduced the variability in x in becoming $\bar{x}$. With prediction intervals we're looking at a single new observation rather than the average, so we have to take the variability of x into account, not just the variability of our estimate of its mean.

Where $SE(Y_* - \hat{Y}_*) = \sqrt{\sigma^2(1 + n^{-1} + S_{XX}^{-1}(X_* - \bar{X})^2)}$

- Prediction intervals are wider or narrower than confidence intervals? Why?

Interpreting different intervals

- Confidence interval: A range of values for the estimated mean response. If we were to take repeated samples we would expect the 95% CIs of the samples to contain the true mean 95% of the time
 - “We are 95% confident the true population value falls within the interval.”
- Prediction interval: A range of values that is likely to contain a single new observation given a specified subset of predictor values.
 - “We are 95% confident that the next observation with these predictor values will fall in this interval.”

How best to predict?

- **GOOD MODEL**
- Parsimony; Overfitting; Collinearity; Assumptions
- Now consider Wage data:
- Are the predictor values realistic? (e.g., age=190?)
- Are populations comparable?
 - Does the model, $\text{wage} \sim \text{year} + \text{jobclass} + \text{race}$, work for northwest region?
 - How about $\text{wage} \sim \text{year} + \text{jobclass} + \text{race}$, for 2010?
 - How about for $\text{wage} \sim \text{factor}(\text{year}) + \text{jobclass} + \text{race}$, for 2010?
- Extreme values and error distribution



Inference

Simple vs. Multiple Regression

$$Y = \beta_0 + \beta_1 X_1 + \epsilon$$

Vs.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

- How does β_1 differ in simple vs. multiple regression?

Simple vs. Multiple Regression

$$Y = \beta_0 + \beta_1 X_1 + \epsilon$$

Vs.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

- How does β_1 differ in simple vs. multiple regression?
- What do we mean when we say “controlling for x” or “holding x constant”?
- Why do we do this?

Prediction vs. Inference

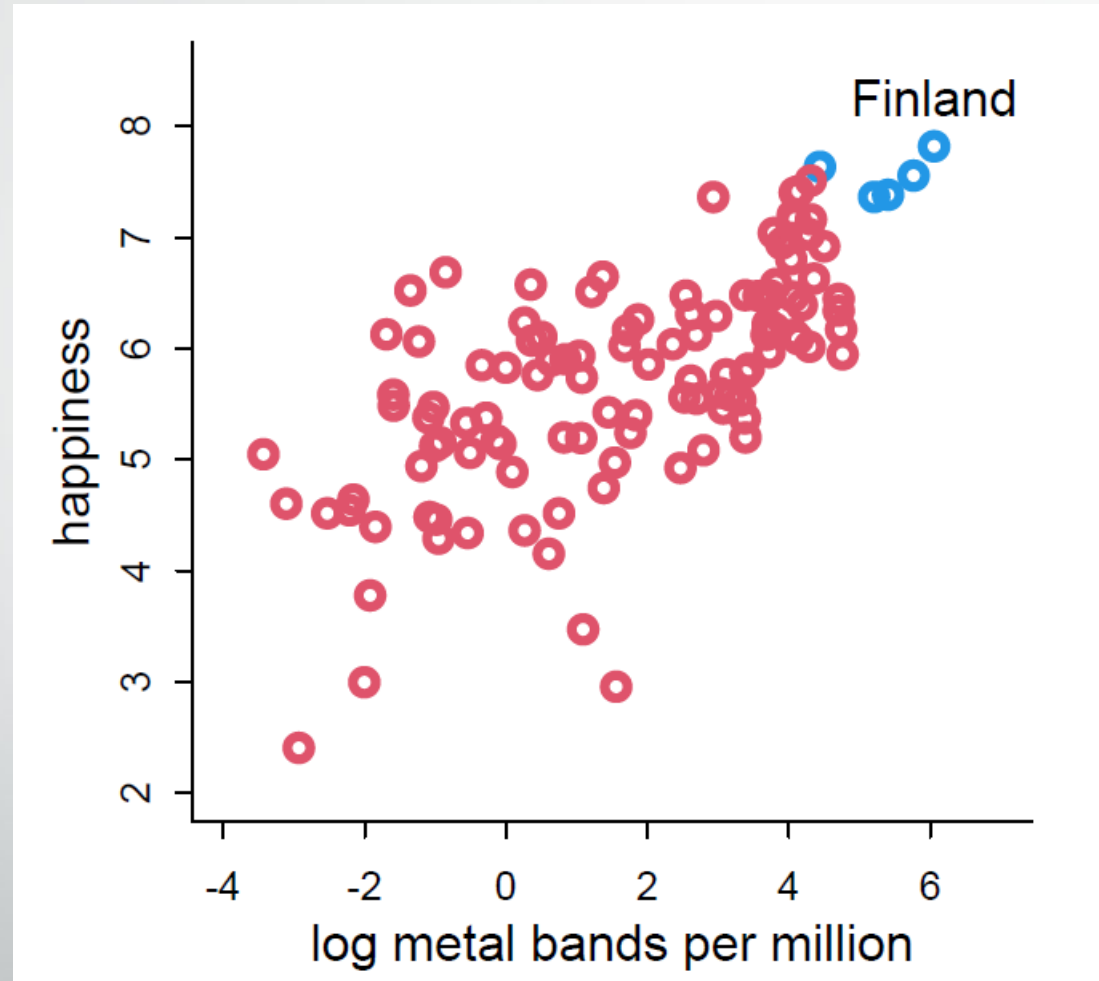
“Prediction is not simple, but it is conceptually easier than explanation.”
(Faraway, p. 59)

- Inference
 - Description
 - Explanation: Causation



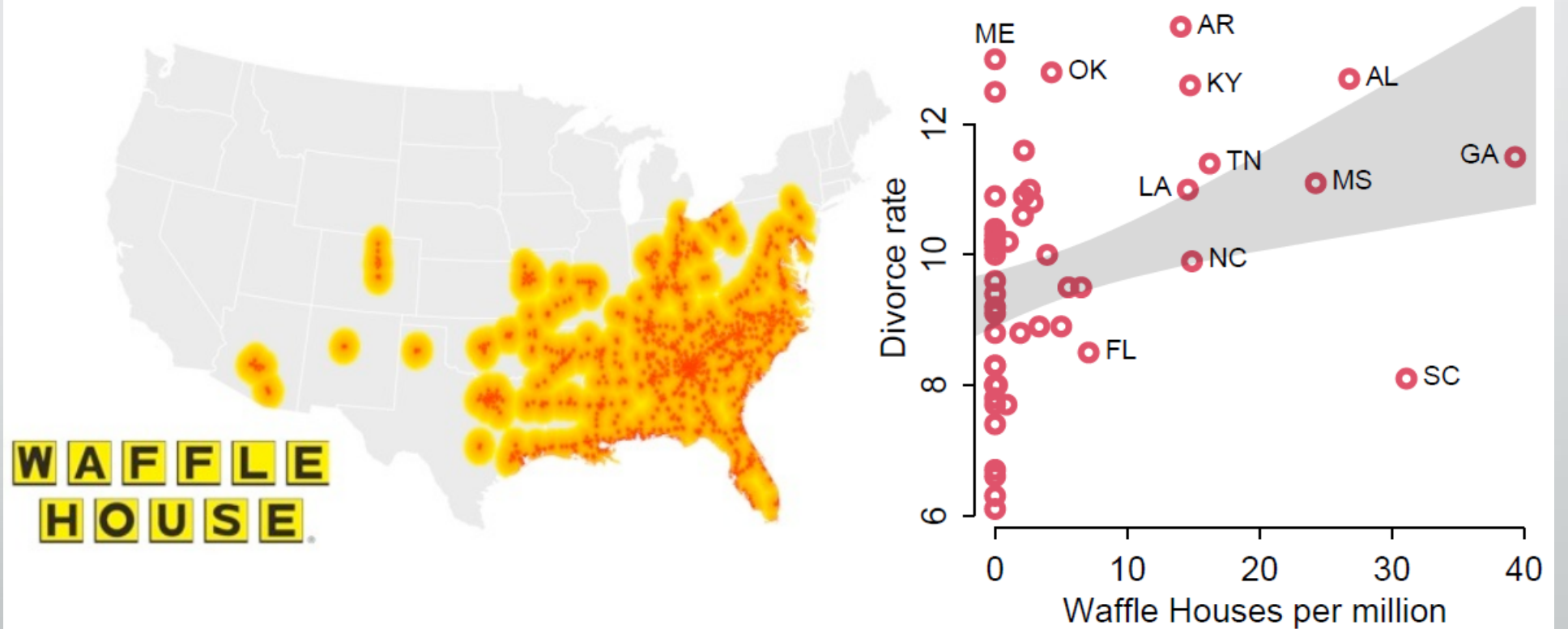
Causation

Causation?

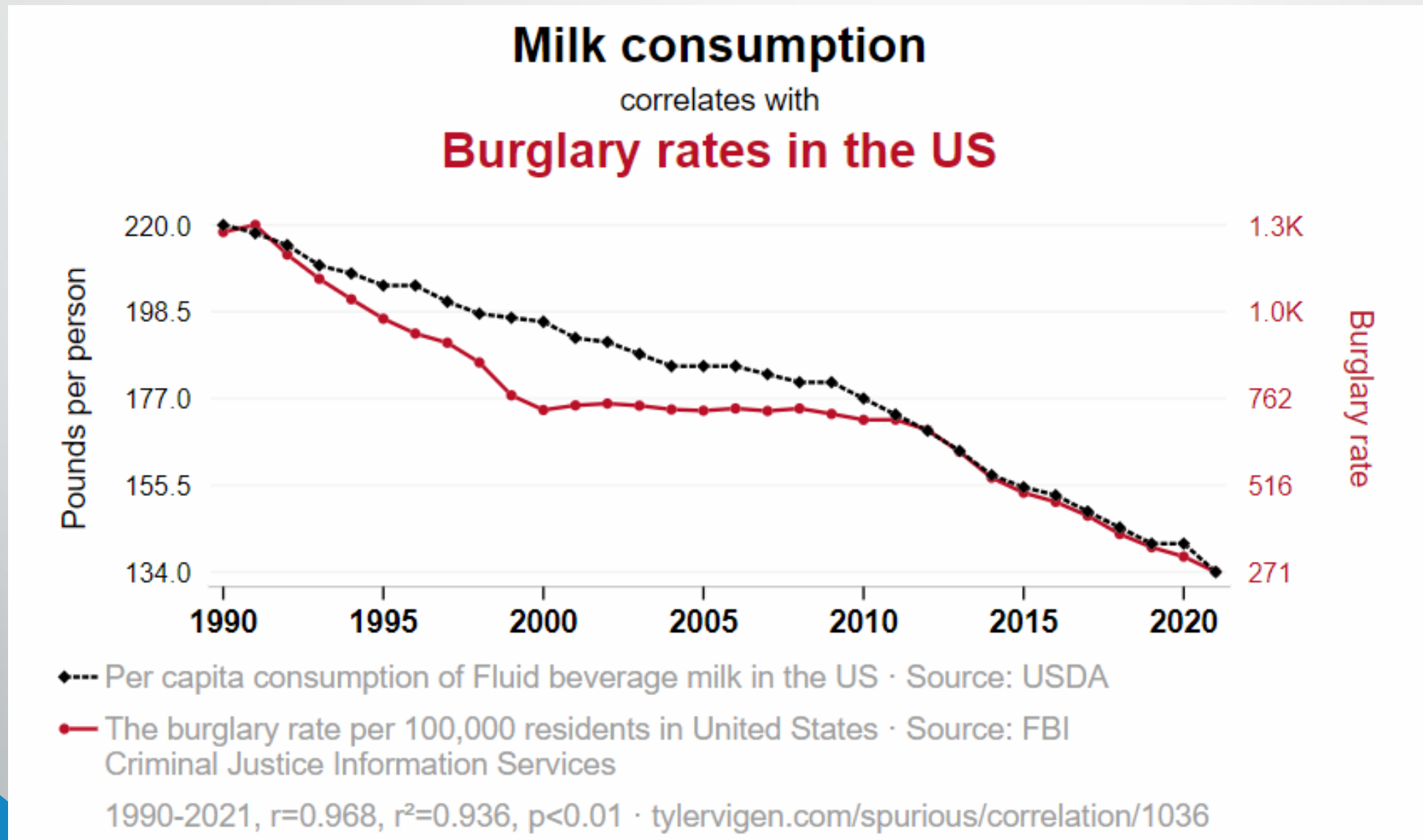


Causation?

Does Waffle House cause divorce?



Correlation is common



Correlation is common

AI explanation

As people stopped crying over spilled milk, they had more time to focus on home security, udderly baffling the burglars. Without moo-ving targets, the thieves were simply milking the clock elsewhere. It seems that dairy deprivation led to a downturn in dairy-ly deeds!



Pounds per person

220.0

198.5

177.0

155.5

134.0

19

◆ Per cap

● The bur

Crimina

1990-20

1.3K

1.0K

762

516

271

Burglary rate

2020

A

tion/1036

What do we mean by causation

- “An antecedent event, condition or characteristic that was necessary for the occurrence of the disease at the moment it occurred, given that other characteristics are fixed.” (Rothman, 2008)
- “We may define a cause to be an object, followed by another...where, if the first object had not been, the second never had existed” (Hume, *An Enquiry Concerning Human Understanding*, 1748)

How do we assess causation?

- The ideal or “gold-standard” would be to conduct a randomized experiment
- Participants are screened for inclusion and then randomly assigned to receive or not receive the treatment/exposure
- They are then observed to see if they develop the outcome of interest

How do we assess causation?

- The primary goal of randomization is to ensure that the 2 groups (i.e., exposed/unexposed, treated/untreated) are “exchangeable”
- Exchangeable means that if those assigned to the unexposed group were instead assigned to the exposed group, we would expect them to have a very similar response as the actual exposed group
- Put another way, any differences between the 2 groups in their response to the exposure/treatment should only be due to random error

Types of Experiments

- Types of Experiments
 - Randomized experiments: An experiment in which units are assigned to receive the treatment by a random process
 - Quasi-experiments: An experiment without random treatment assignment
 - Natural experiments: Not really an experiment; the cause cannot be manipulated; a study that contrasts a naturally occurring event (e.g., natural disaster) with a comparison population
- Why is randomization beneficial?
- Why not always use randomization?

Why don't we only use random experiments?

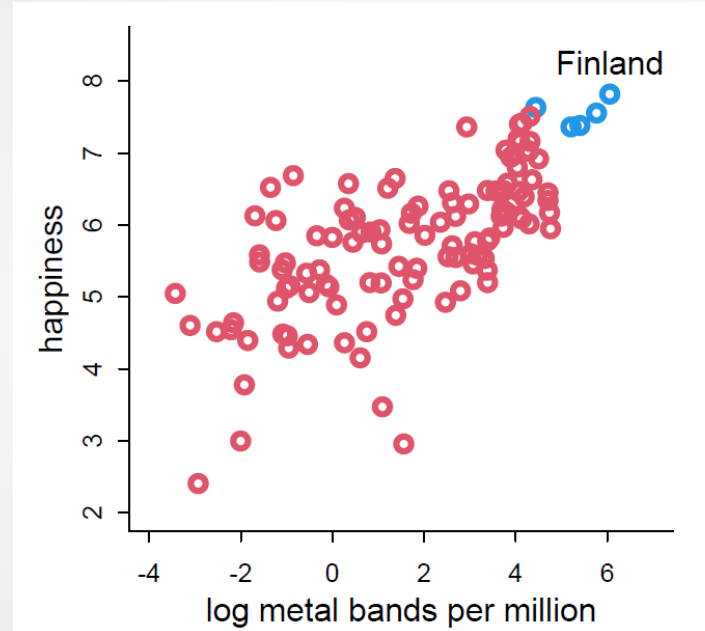
- Not possible: Some things can't be randomized (e.g., genetics)
- Not ethical: Can't purposely expose people to something harmful
- Expensive
- Challenging with rare outcomes

Observational studies

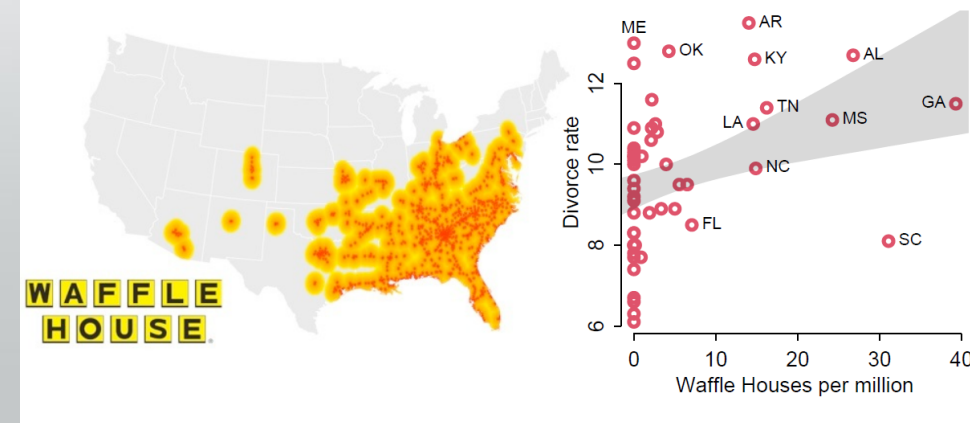
- Cohort studies
- Case-control studies
- Cross-sectional studies
- Ecological studies
- Among others...

Causal inference with observational data

- How do we try to ensure that our models are not simply capturing spurious correlations?



Does Waffle House cause divorce?





Hill's Viewpoints

Hill's Viewpoints

1. Strength of association
2. Consistency
3. Specificity
4. Temporality
5. Biological gradient
6. Plausibility
7. Coherence
8. Experimental evidence
9. Analogy

Proposed in 1965 by English statistician Sir Austin Bradford Hill, Hill's viewpoints represent an early effort to formalize how we approach establishing whether an observed relationship is causal.

Not all have held up to scrutiny in all situations.

Hill's Viewpoints

1. Strength of association
2. Consistency
3. Specificity
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A weak association doesn't mean that a relationship is not causal, but a strong relationship is more likely to be causal.

Doll and Hill's study on cigarette smoking and lung cancer.

Hill's Viewpoints

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← An association is repeatedly observed in different populations and under different circumstances.

Subject to publication bias (i.e., non-null results more likely to be published). Can also be impacted by confounding or other biases.

Hill's Viewpoints

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Cause leads to a single effect, not multiple.

Hill's Viewpoints

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****This point is invalid****

effect, not

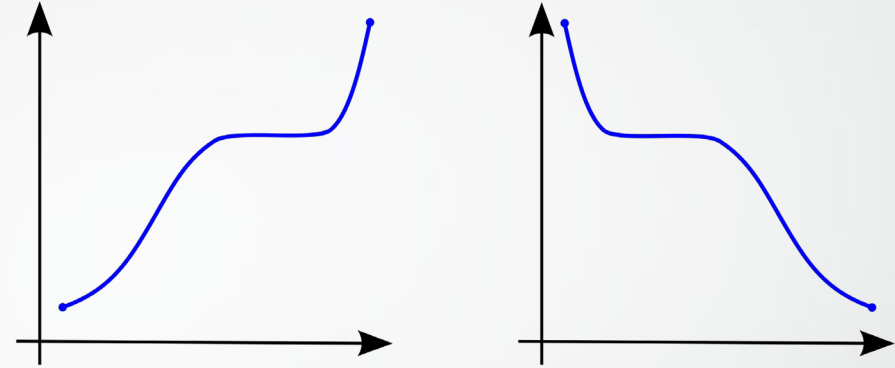
Hill's Viewpoints

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Exposure must precede outcome.

Hill's Viewpoints

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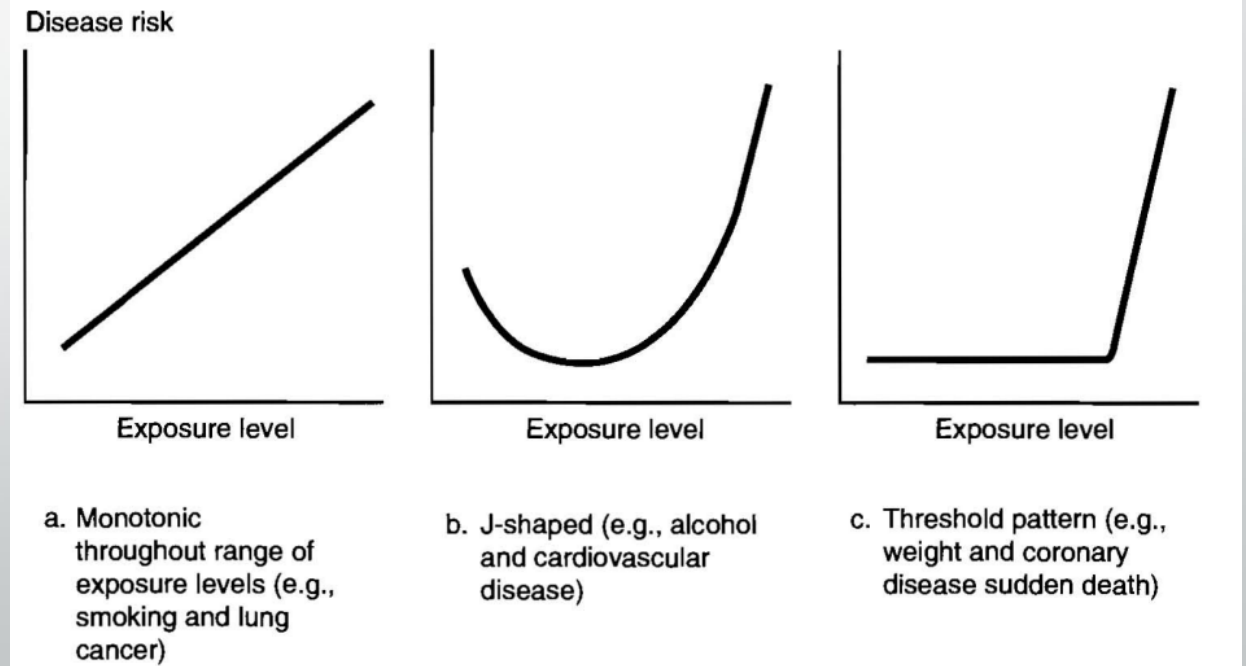


A monotonic relationship between level of exposure and outcome is considered evidence of causality.

Hill's Viewpoints

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Not always the case, however, as many other shapes possible.



Hill's Viewpoints

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Must be consistent with our understanding of biology

Hill's Viewpoints

1. Strength of association
2. Consistency
3. Specificity
4. Temporality
5. Biological plausibility
6. Plausibility
7. Coherence
8. Experimental evidence
9. Analogy

Our understanding of biology and other sciences is constantly growing. A relationship may not be consistent with our current understanding, but can still be accurate.

Hill's Viewpoints

1. Strength of association
2. Consistency
3. Specificity
4. Temporality
5. Biological gradient
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“...the cause-and-effect story should make sense with all knowledge available to the researcher...” Fedak et al., 2015

Hill's Viewpoints

1. Strength of association
2. Consistency
3. Specificity
4. Temporality
5. Biological gradient
6. Plausibility
7. Coherence ←
8. Experimental evidence
9. Analogy

Cannot rule out causal relationship as it relies on the knowledge currently available which might not be sufficient.

knowledge available to the researcher..." Fedak et al., 2015

Hill's Viewpoints

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2. Consistency
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Presence of experimental evidence extremely helpful.

Hill's Viewpoints

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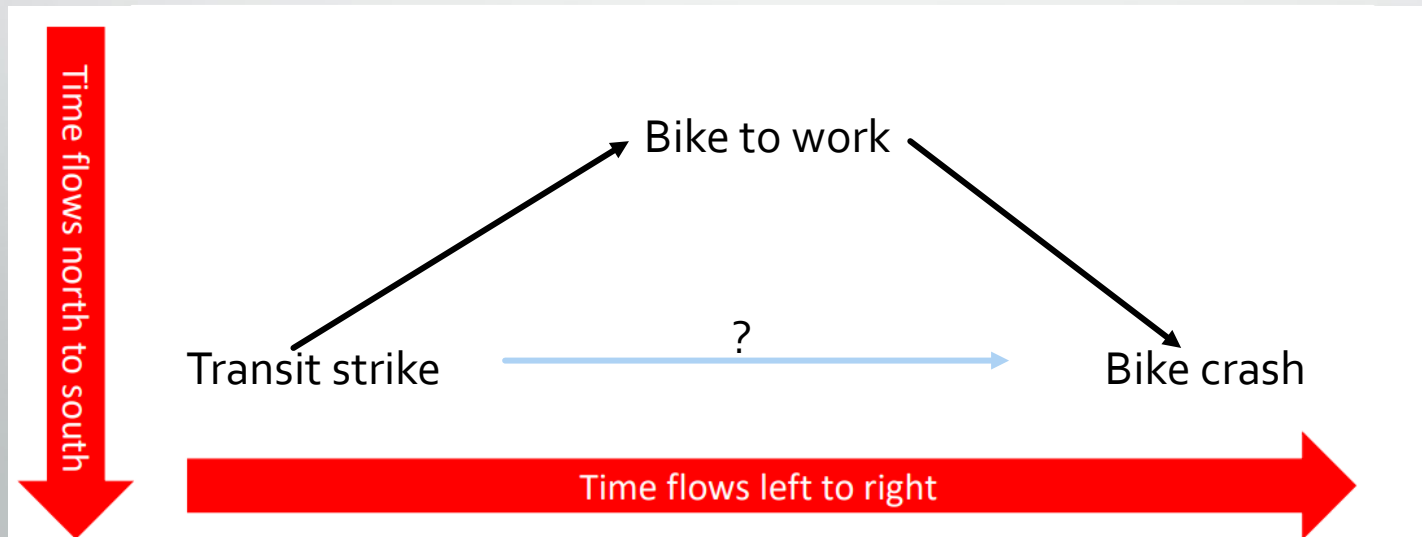
← The effect of similar factors may be considered.



DAGs

Directed Acyclic Graphs (DAGs)

- DAGs are another tool that is commonly used by researchers to conceptualize the relationship being modeled
- The presence of an arrow indicates the possibility/assumption of a relationship, not that it absolutely exists
 - This assumption should be based on prior evidence!



What can DAGs tell us?

Ye Olde Causal Alchemy

The Four Elemental Confounds



The Fork

$X \leftarrow Z \rightarrow Y$

The Pipe

$X \rightarrow Z \rightarrow Y$



The Collider

$X \rightarrow Z \leftarrow Y$

The Descendant

$X \rightarrow Z \rightarrow Y$

$\downarrow$
A

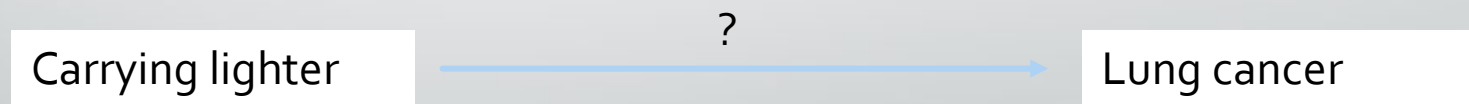




Confounders, mediators, colliders, and
descendents

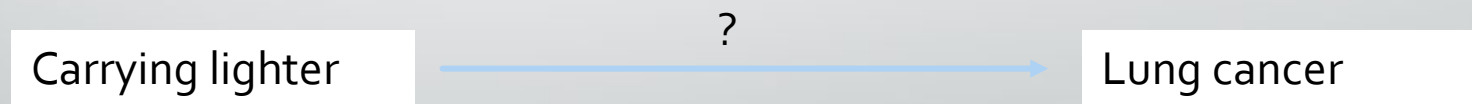
The Fork (AKA confounding)

- Suppose we were interested in whether carrying matches placed you at greater risk of lung cancer.



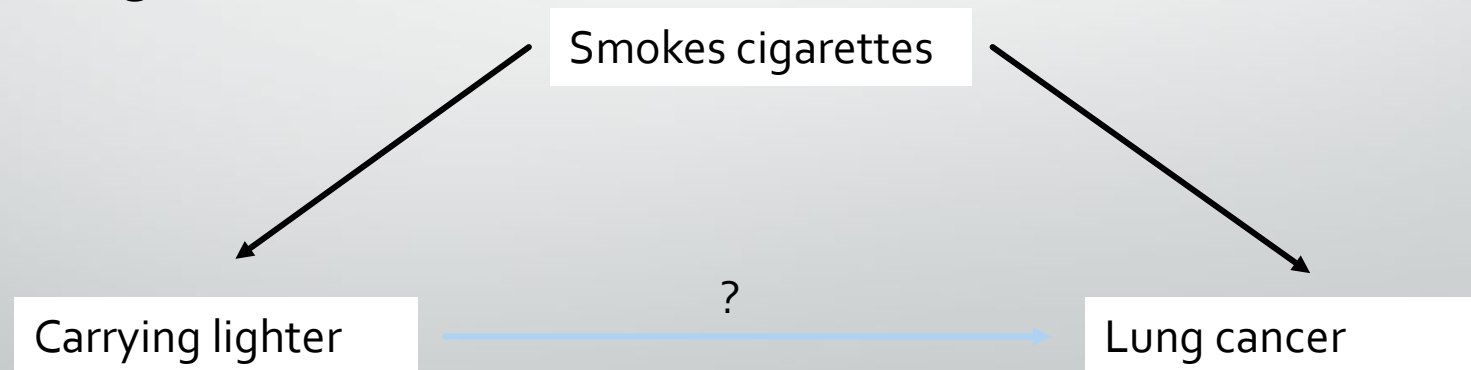
The Fork (AKA confounding)

- Suppose we were interested in whether carrying matches placed you at greater risk of lung cancer.
- What might be missing from this DAG?



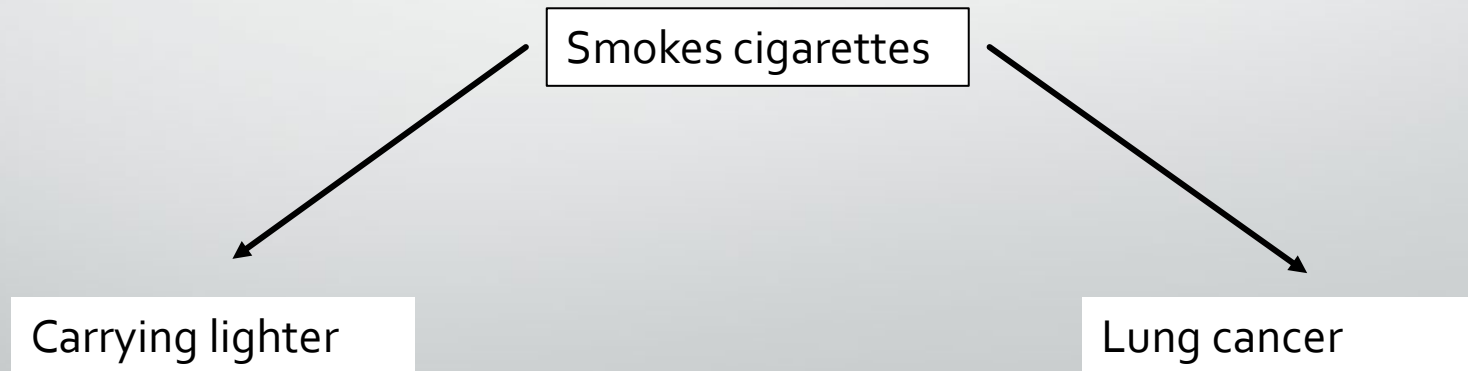
The Fork (AKA confounding)

- Suppose we were interested in whether carrying matches placed you at greater risk of lung cancer.
- Cigarette smoking is a confounder of the “relationship” between carrying a lighter and lung cancer



What to do?

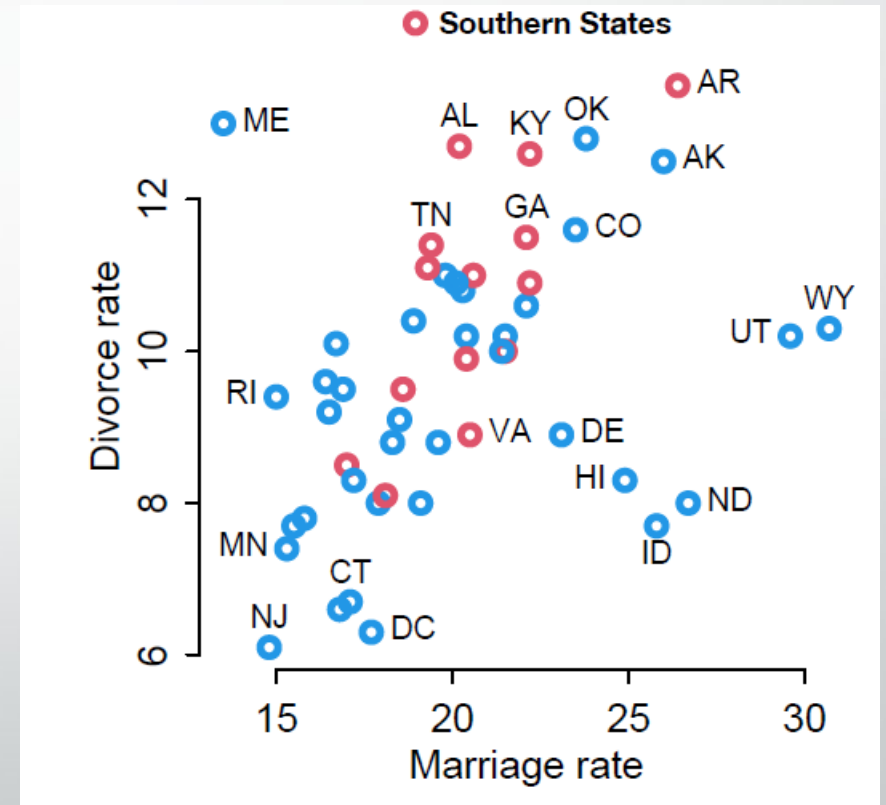
- Multiple regression
 - Can control for potential confounders



Addressing confounding example

- Why do regions of the US with higher rates of marriage also have higher rates of divorce?
- We will use the “WaffleDivorce.csv” dataset (posted on Canvas) to explore the divorce rate as a function of the marriage rate.
 - Note this file used “;” as a delimiter so you need to use the following code to import to R:

```
read.csv2("C:/Users/jkubale/Downloads/WaffleDivorce.csv", sep=";")
```



Addressing confounding example

- Let's fit a model (starting simple):

```
mod1 <- lm(Divorce ~ as.numeric(Marriage), waffle)
```

```
> summary(mod1)

Call:
lm(formula = Divorce ~ as.numeric(Marriage), data = waffle)

Residuals:
    Min       1Q   Median       3Q      Max
-3.0068 -1.2173  0.1214  1.1805  4.4971

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    6.08404    1.31337   4.632 2.78e-05 ***
as.numeric(Marriage) 0.17918    0.06418   2.792  0.00751 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.706 on 48 degrees of freedom
Multiple R-squared:  0.1397,    Adjusted R-squared:  0.1218
F-statistic: 7.793 on 1 and 48 DF,  p-value: 0.007507

> |
```

Addressing confounding example

- Let's fit a model (starting simple):

```
mod1 <- lm(Divorce ~ as.numeric(Marriage), waffle)
```

Suggests that with higher marriage rates we would expect higher rates of divorce.

But is that the whole story?

```
> summary(mod1)

Call:
lm(formula = Divorce ~ as.numeric(Marriage), data = waffle)

Residuals:
    Min       1Q   Median       3Q      Max
-3.0068 -1.2173  0.1214  1.1805  4.4971

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    6.08404    1.31337   4.632 2.78e-05 ***
as.numeric(Marriage) 0.17918    0.06418   2.792  0.00751 **
---
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F-statistic: 7.793 on 1 and 48 DF,  p-value: 0.007507

> |
```

Addressing confounding example

- Let's fit a model (starting simple):

```
mod1 <- lm(Divorce ~ as.numeric(Marriage), waffle)
```

Might the age at which people tend to get married impact their likelihood of divorce?

Suggests that with higher marriage rates we would expect higher rates of divorce.

But is that the whole story?

```
Residuals:
    Min       1Q   Median       3Q      Max
-3.0068 -1.2173  0.1214  1.1805  4.4971

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    6.08404    1.31337   4.632 2.78e-05 ***
as.numeric(Marriage) 0.17918    0.06418   2.792  0.00751 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.706 on 48 degrees of freedom
Multiple R-squared:  0.1397,    Adjusted R-squared:  0.1218
F-statistic: 7.793 on 1 and 48 DF,  p-value: 0.007507

> |
```

Addressing confounding example

- Let's fit another model this time with both Marriage and MedianAgeMarriage as predictors

```
mod2 <- lm(Divorce ~ as.numeric(Marriage)  
as.numeric(MedianAgeMarriage), waffle)
```

```
> summary(mod2)  
  
Call:  
lm(formula = Divorce ~ as.numeric(Marriage) + as.numeric(MedianAgeMarriage),  
    data = waffle)  
  
Residuals:  
    Min       1Q   Median       3Q      Max   
-4.5177 -0.9828 -0.0458  0.9224  3.2818   
  
Coefficients:  
                Estimate Std. Error t value Pr(>|t|)      
(Intercept)      36.87665     7.66104   4.814 1.58e-05 ***  
as.numeric(Marriage) -0.05686     0.08053  -0.706 0.483594   
as.numeric(MedianAgeMarriage) -0.99965     0.24593  -4.065 0.000182 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 1.483 on 47 degrees of freedom  
Multiple R-squared:  0.3634,    Adjusted R-squared:  0.3364   
F-statistic: 13.42 on 2 and 47 DF,  p-value: 2.455e-05  
  
> |
```

Addressing confounding example

- Let's fit another model this time with both Marriage and MedianAgeMarriage as predictors

```
mod2 <- lm(Divorce ~ as.numeric(Marriage)  
as.numeric(MedianAgeMarriage), waffle)
```

As median age at marriage increases the divorce rate decreases.

What happened to the coefficient for Marriage?

```
> summary(mod2)  
  
Call:  
lm(formula = Divorce ~ as.numeric(Marriage) + as.numeric(MedianAgeMarriage),  
    data = waffle)  
  
Residuals:  
    Min       1Q   Median       3Q      Max   
-4.5177 -0.9828 -0.0458  0.9224  3.2818   
  
Coefficients:  
                Estimate Std. Error t value Pr(>|t|)      
(Intercept)      36.87665     7.66104   4.814 1.58e-05 ***  
as.numeric(Marriage) -0.05686     0.08053  -0.706 0.483594   
as.numeric(MedianAgeMarriage) -0.99965     0.24593  -4.065 0.000182 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 1.483 on 47 degrees of freedom  
Multiple R-squared:  0.3634,    Adjusted R-squared:  0.3364   
F-statistic: 13.42 on 2 and 47 DF,  p-value: 2.455e-05  
  
> |
```



Times we don't want to include
covariates

When we don't want to include covariates

Ye Olde Causal Alchemy

The Four Elemental Confounds



The Fork

$X \leftarrow Z \rightarrow Y$

The Pipe

$X \rightarrow Z \rightarrow Y$



The Collider

$X \rightarrow Z \leftarrow Y$

The Descendant

$X \rightarrow Z \rightarrow Y$

$\downarrow$
A



When we don't want to include covariates

Ye Olde Causal Alchemy The Four Elemental Confounds

Some exceptions with the pipe
(i.e., interested in direct vs. total
effect) though even these cases
can be problematic.



The Collider

$X \longrightarrow Z \longleftarrow Y$

The Descendant

$X \longrightarrow Z \longrightarrow Y$
 $\downarrow$
 A



The Pipe (AKA mediation)

The Pipe

$$X \rightarrow Z \rightarrow Y$$

Z is a “mediator”

X and Y are associated

$$Y \not\perp\!\!\!\perp X$$

Influence of X on Y transmitted through Z

Once stratified by Z, no association

$$Y \perp\!\!\!\perp X \mid Z$$

The Pipe (AKA mediation)

$$X \rightarrow Z \rightarrow Y$$

```
n <- 1000  
X <- rbern( n , 0.5)  
Z <- rbern( n , (1-X)*0.1 + X*0.9 )  
Y <- rbern( n , (1-Z)*0.1 + Z*0.9 )
```

	Y	
X	0	1
0	430	87
1	93	390

$Y \nparallel X$

```
> cor(X,Y)  
[1] 0.64
```

The Pipe (AKA mediation)

$$X \rightarrow Z \rightarrow Y$$

```
n <- 1000
X <- rbern( n , 0.5)
Z <- rbern( n , (1-X)*0.1 + X*0.9 )
Y <- rbern( n , (1-Z)*0.1 + Z*0.9 )
```

	Y	
X	0	1
0	430	87
1	93	390

$Y \not\perp\!\!\!\perp X$

```
> cor(X,Y)
[1] 0.64
```

	Y	
X	0	1
0	422	39
1	53	5

	Y	
X	0	1
0	8	48
1	40	385

$Y \perp\!\!\!\perp X \mid Z$

```
> cor(X[Z==0],Y[Z==0])
[1] 0.002
> cor(X[Z==1],Y[Z==1])
[1] 0.052
```

The Pipe (AKA mediation)

$$X \rightarrow Z \rightarrow Y$$

```
cols <- c(4,2)

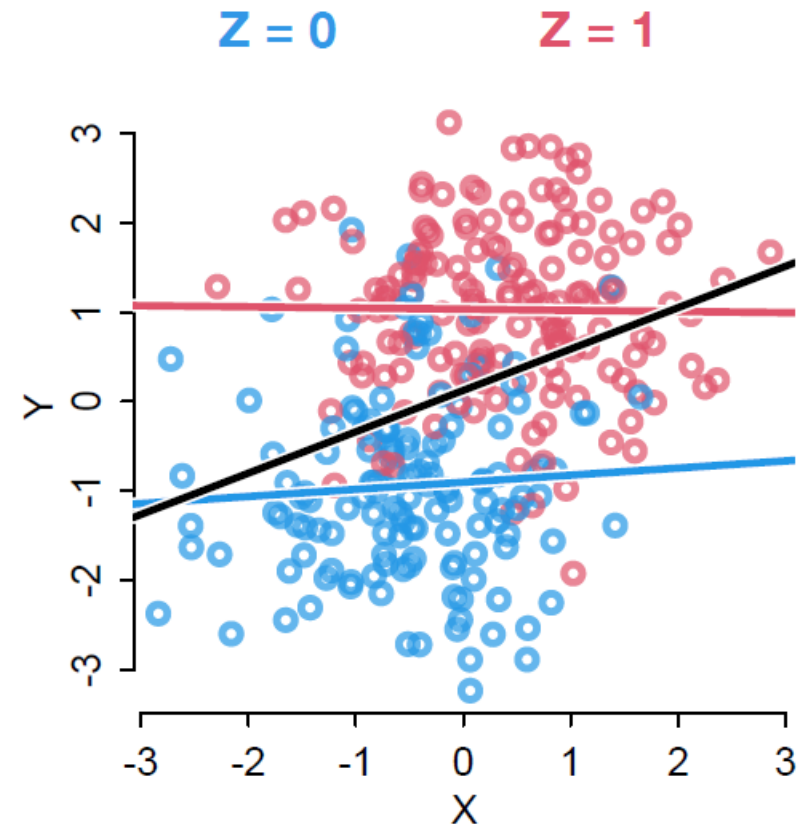
N <- 300
X <- rnorm(N)
Z <- rbern(N,inv_logit(X))
Y <- rnorm(N,(2*Z-1))

plot( X , Y , col=cols[Z+1] , lwd=3 )

abline(lm(Y[Z==1]~X[Z==1]),col=2,lwd=3)

abline(lm(Y[Z==0]~X[Z==0]),col=4,lwd=3)

abline(lm(Y~X),lwd=3)
```



The Collider

The Collider

$$X \rightarrow Z \leftarrow Y$$

Z is a “collider”

X and Y are not associated (share no causes)

$$Y \perp\!\!\!\perp X$$

X and Y both influence Z

Once stratified by Z , X and Y associated

$$Y \not\perp\!\!\!\perp X \mid Z$$

The Collider

$$X \rightarrow Z \leftarrow Y$$

```
n <- 1000  
X <- rbern( n , 0.5 )  
Y <- rbern( n , 0.5 )  
Z <- rbern( n , ifelse(X+Y>0,0.9,0.2) )
```

X	Y	
	0	1
0	243	236
1	250	271

$$Y \perp\!\!\!\perp X$$

```
> cor(X,Y)  
[1] 0.027
```


The Collider

$$X \rightarrow Z \leftarrow Y$$

```
n <- 1000
X <- rbern( n , 0.5 )
Y <- rbern( n , 0.5 )
Z <- rbern( n , ifelse(X+Y>0,0.9,0.2) )
```

	Y	
X	0	1
0	243	236
1	250	271

$Y \perp\!\!\!\perp X$

```
> cor(X,Y)
[1] 0.027
```

	Y	
X	0	1
0	200	19
1	32	29

	Y	
X	0	1
0	43	217
1	218	242

$Y \not\perp\!\!\!\perp X | Z$

```
> cor(X[Z==0],Y[Z==0])
[1] 0.43
> cor(X[Z==1],Y[Z==1])
[1] -0.31
```

The Collider

$$X \rightarrow Z \leftarrow Y$$

```
cols <- c(4,2)

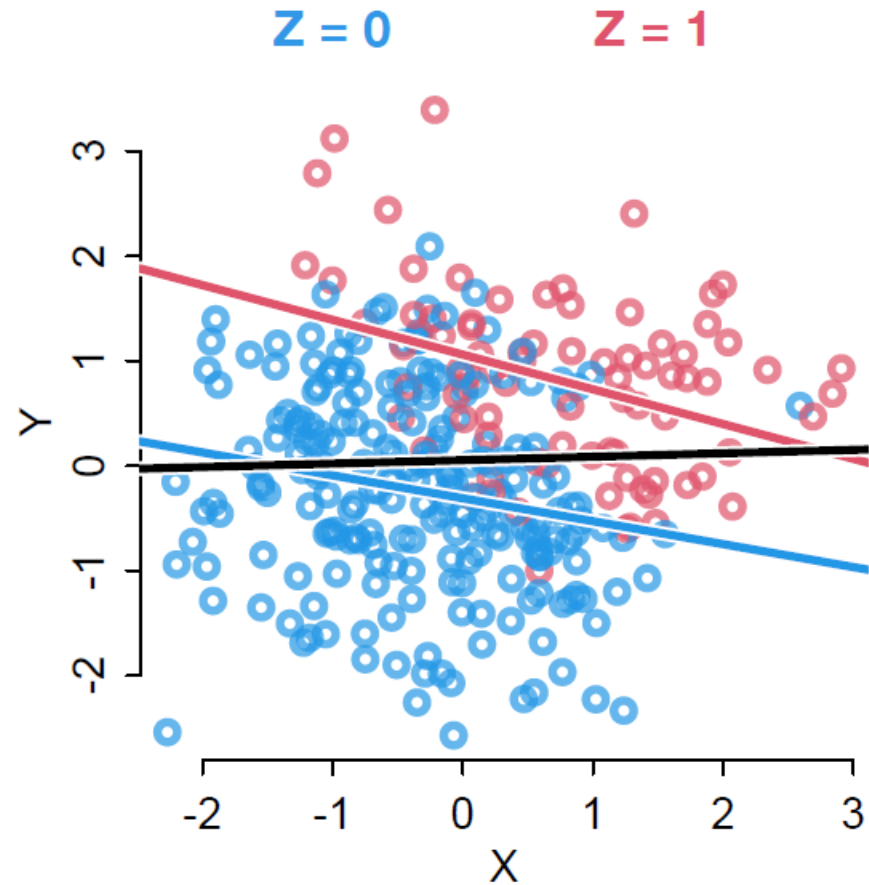
N <- 300
X <- rnorm(N)
Y <- rnorm(N)
Z <- rbern(N,inv_logit(2*X+2*Y-2))

plot( X , Y , col=cols[Z+1] , lwd=3 )

abline(lm(Y[Z==1]~X[Z==1]),col=2,lwd=3)

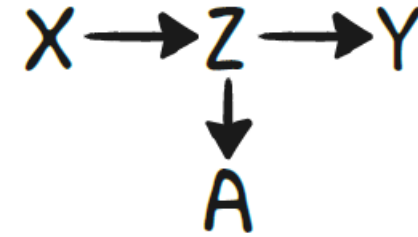
abline(lm(Y[Z==0]~X[Z==0]),col=4,lwd=3)

abline(lm(Y~X),lwd=3)
```



The Descendant

The Descendant



A is a “descendant”

- Depends on what it descends from.
 - E.g., If it descends from a mediator, controlling for it in the model will result in similar behavior as if you included the mediator itself.